MEDIUM ADVANCED MATH

Advanced Math

13

10 questions · ~15 min

SAT QUESTION BANK 2026-05-29

Q1 ADVANCED MATH

$$14j + 5k = m$$

The given equation relates the numbers j, k, and m. Which equation correctly expresses k in terms of j and m?

A. $k = \frac{m - 14j}{5}$

B. $k = \frac{1}{5}m - 14j$

C. $k = \frac{14j - m}{5}$

D. $k = 5m - 14j$

Q2

GRID-IN ADVANCED MATH

A rubber ball bounces upward one-half the height that it falls each time it hits the ground. If the ball was originally dropped from a distance of 20.0 feet above the ground, what was its maximum height above the ground, in feet, between the third and fourth time it hit the ground?

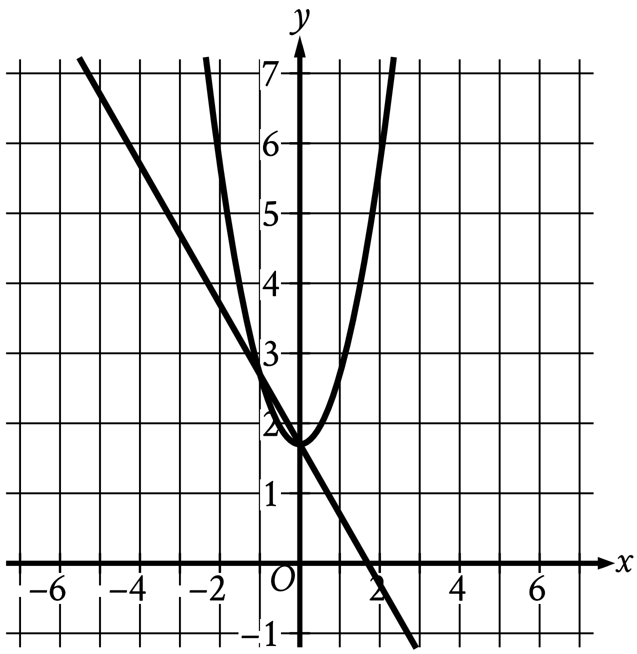
STUDENT-PRODUCED RESPONSE

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$y = x^2 + 1.7$$

$$y = 1.7 - x$$

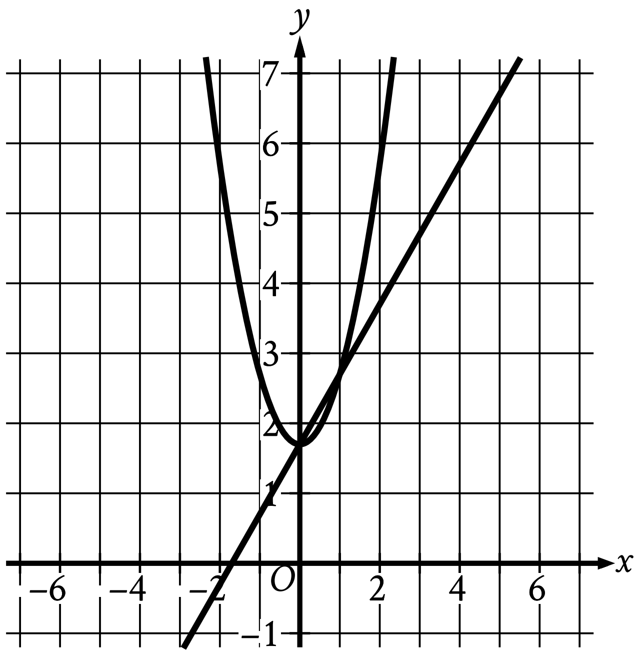
Which graph represents the given system of equations?



A.

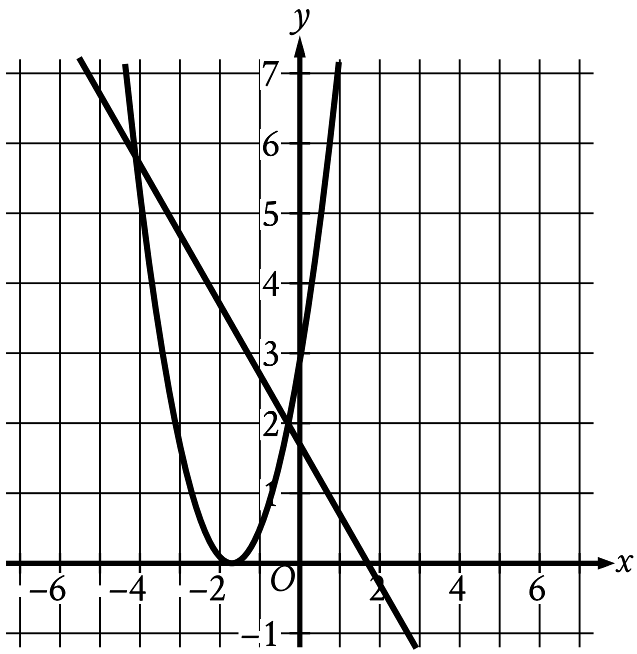
- For the line in the system:
  - The line slants sharply down from left to right.
  - The line passes through the following points:
    - (negative 1 comma 2.7)
    - (0 comma 1.7)
- For the parabola in the system:
  - The parabola opens upward.
  - The vertex is at point (0 comma 1.7).
  - The parabola passes through the following points:
    - (negative 1 comma 2.7)
    - (0 comma 1.7)
    - (1 comma 2.7)

B.



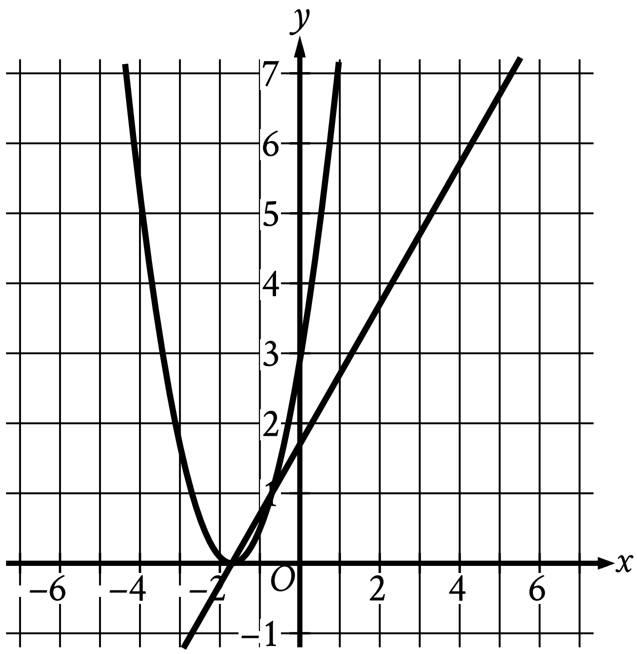
- For the line in the system:
  - The line slants sharply up from left to right.
  - The line passes through the following points:
    - (0 comma 1.7)
    - (1 comma 2.7)
- For the parabola in the system:
  - The parabola opens upward.
  - The vertex is at point (0 comma 1.7).
  - The parabola passes through the following points:
    - (negative 1 comma 2.7)
    - (0 comma 1.7)
    - (1 comma 2.7)

C.



- For the line in the system:
    - The line slants sharply down from left to right.
    - The line passes through the following points:
        - (negative 1 comma 2.7)
        - (0 comma 1.7)
- For the parabola in the system:
    - The parabola opens upward.
    - The vertex is at point (negative 1.7 comma 0).
    - The parabola passes through the following points:
        - (negative 2.7 comma 1)
        - (negative 1.7 comma 0)
        - (negative 0.7 comma 1)

D.



- For the line in the system:
  - The line slants sharply up from left to right.
  - The line passes through the following points:
    - (0 comma 1.7)
    - (1 comma 2.7)
- For the parabola in the system:
  - The parabola opens upward.
  - The vertex is at point (negative 1.7 comma 0).
  - The parabola passes through the following points:
    - (negative 2.7 comma 1)
    - (negative 1.7 comma 0)
    - (negative 0.7 comma 1)

$$mt = -0.0274\frac{t}{7}^2 + 7.3873\frac{t}{7} + 75.032$$

The function m gives the predicted body mass mt, in kilograms kg, of a certain animal t days after it was born in a wildlife reserve, where $t \leq 390$. Which of the following is the best interpretation of the statement “$m330$ is approximately equal to 362” in this context?

A. The predicted body mass of the animal was approximately 330 kg 362 days after it was born.

B. The predicted body mass of the animal was approximately 362 kg 330 days after it was born.

C. The predicted body mass of the animal was approximately 362 kg $\frac{330}{7}$ days after it was born.

D. The predicted body mass of the animal was approximately $\frac{330}{7}$ kg 362 days after it was born.

Q5

ADVANCED MATH

$$x^2 - x - 1 = 0$$

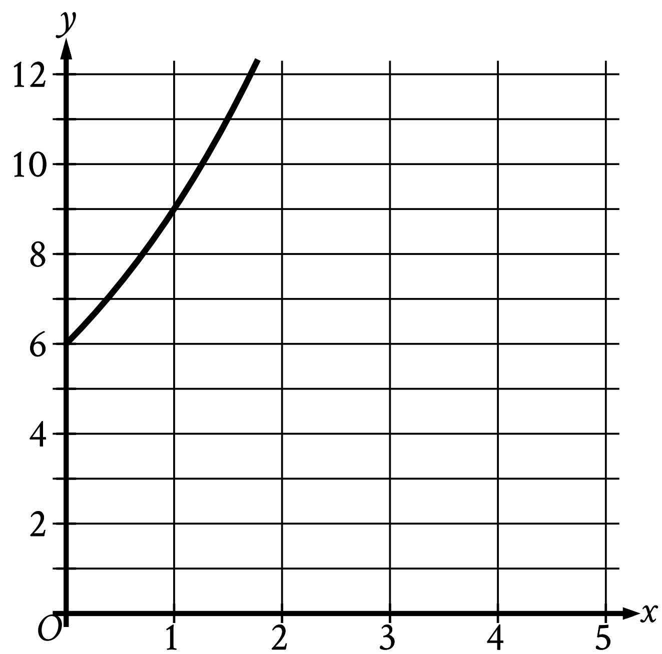
What values satisfy the equation above?

A. $x = 1$ and $x = 2$

B. $x = -\frac{1}{2}$ and $x = \frac{3}{2}$

C. $x = \frac{1+\sqrt{5}}{2}$ and $x = \frac{1-\sqrt{5}}{2}$

D. $x = \frac{-1+\sqrt{5}}{2}$ and $x = \frac{-1-\sqrt{5}}{2}$



- Moving from left to right:
  - The curve passes through quadrant 1.
  - The curve trends up sharply.
- The curve passes through the following points:
  - (0 comma 6)
  - (1 comma 9.0)

The graph gives the estimated population y, in thousands, of a town x years since 2003, where $0 \leq x \leq 5$. Which of the following best describes the increase in the estimated population from $x = 0$ to $x = 1$?

A. The estimated population at $x = 1$ is 0.5 times the estimated population at $x = 0$.

B. The estimated population at $x = 1$ is 1.5 times the estimated population at $x = 0$.

C. The estimated population at $x = 1$ is 2.5 times the estimated population at $x = 0$.

D. The estimated population at $x = 1$ is 3.5 times the estimated population at $x = 0$.

Q7 GRID-IN ADVANCED MATH

$$x + y = 17$$
$$xy = 72$$

If one solution to the system of equations above is (x, y), what is one possible value of x ?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q8 ADVANCED MATH

A sample of a certain isotope takes 29 years to decay to half its original mass. The function $st = 184(0.5)^{\frac{t}{29}}$ gives the approximate mass of this isotope, in grams, that remains t years after a 184-gram sample starts to decay. Which statement is the best interpretation of $s87 = 23$ in this context?

A. Approximately 23 grams of the sample remains 87 years after the sample starts to decay.

B. The mass of the sample has decreased by approximately 23 grams 87 years after the sample starts to decay.

C. The mass of the sample has decreased by approximately 87 grams 23 years after the sample starts to decay.

D. Approximately 87 grams of the sample remains 23 years after the sample starts to decay.

Q9

ADVANCED MATH

If $\frac{42}{x} = 7x$, what is the value of $7x^2$?

A. 6

B. 7

C. 42

D. 294

Q10

GRID-IN ADVANCED MATH

The function f is defined by $f x = 8x^3 + 4$. What is the value of $f 2$?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.